Jeonghan **Kim**

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Research Interests

My research interests focus on robot path and motion planning for high-dimensional, constraint-rich environments. I aim to research real-time planning and control methods related to optimization-based planning, diffusion-based planners, and constraint-aware approaches under equality and inequality constraints.

Education

Hanyang University *Ansan, South Korea*

B.S. in Robot Engineering, minor in Electronic Engineering *Mar. 2018 - Feb. 2024*

- GPA: 3.91 / 4.5; Major GPA: 4.13 / 4.5

Professional Experience

Roboe Technologies *Seoul, South Korea*

Software Engineer, Motion Algorithm Team *Mar. 2024 - Present*

- Developed reachability-based workspace analysis and optimization tooling for conveyor-mounted mobile robots.
- Implemented geometry-based conveyor jamming detection and optimized PointCloud2 filtering to operate at 20 Hz.
- Built velocity-based predictive collision shielding with MoveIt2 FCL collision detection.
- Reconstructed motion-task execution from Behavior Tree logic into a graph-based architecture and reduced Python GIL bottlenecks with Ray-based multiprocessing.
- Implemented C++ motion planning algorithms for moveL and moveJ trajectories in virtual controllers.

Gole Robotics *Seoul, South Korea*

Intern, Autonomous Driving Software *Aug. 2023 - Oct. 2023*

- Implemented a packet-based serial communication library aligned with embedded controller protocols.
- Benchmarked IMU, LiDAR, ToF, and barometer candidates using mean/variance analysis and PlotJuggler to support sensor selection.
- Standardized the team development environment with Docker, Docker Compose, VS Code, Git, ROS, and ROS2.
- Contributed autonomous mobile robot software to a company product later recognized as a CES 2024 Innovation Awards Honoree.

Research Experience

RAISE Lab (Prof. Youngmoon Lee) *Hanyang University*

Undergraduate Researcher *Jan. 2023 - Feb. 2024*

- Conducted undergraduate research on manipulator-based HRI systems, focusing on robot motion planning, control, and system integration.
- Developed perception and planning modules for a mobile-robot patrol project under Prof. Youngmoon Lee's supervision.

Publications

Snapbot: Enabling Dynamic Human–Robot Interactions for Real-Time Computational Photography *ACM/IEEE HRI 2024, Boulder, USA*

C. Choi, J. Kim, Y. Nam, and Y. Lee

- Developed an end-to-end HRI system integrating perception, control, and computational photography for dynamic camera composition.
- Designed a control loop that minimizes the angular difference between human head pose and the robot TCP direction using head pose estimation.
- Implemented manipulator velocity control based on pseudo-inverse Jacobian formulation for smooth camera motion.

Separable Integrated Drone–Mobile Robot System for Unmanned Patrol *ICROS 2023, Samcheok, South Korea*

J. Kim, J. Park, H. Lee, S. Park, Y. Nam, G. Kwon, W. Jung, and Y. Lee

- Proposed a cooperative drone–mobile robot platform enabling energy-efficient patrol and terrain coverage.
- *Paper received the Undergraduate Best Paper Award, ICROS 2023.*

Selected Projects

Planning - Python 3D Path Planning Library *GitHub (Open Source)*

Creator & Main Contributor *Sep. 2025 - Present*

- Built a Python path-planning library with a unified planner architecture and Viser-based visualization for RRT, RRT-Connect, RRG, RRT*, Informed RRT*, PRM, and PRM*.
- Extended the library with N-dimensional planning support, terrain-aware Riemannian metric costs, and diffusion-based trajectory planning examples.

Snapbot: Computational Photography and HRI Robot *Hanyang University*

Manipulator Control & Perception *Sep. 2023 - Dec. 2023*

- Developed a manipulator-based computational photography robot that tracks people and selects camera poses for dynamic human-robot interaction.
- Built a YOLO/depth-based human perception pipeline to estimate the relative TF between the robot and a detected person.
- Implemented inverse-kinematics closed-loop manipulator control with filtering and damped pseudo-inverse singularity avoidance.
- Contributed to system implementation and experiments for KROC 2024 and ACM/IEEE HRI 2024 papers.

Separable Drone-Mobile Robot System for Unmanned Patrol *Hanyang University*

Project Lead *Jan. 2023 - Jun. 2023*

- Led development of an integrated drone-mobile robot system where a ground robot carries, charges, and deploys a drone for last-mile patrol over rugged terrain.
- Designed ROS architecture, indoor/outdoor autonomous operation, app-to-ROS interface, and point-cloud based terrain perception modules.
- Fused IMU, GNSS, and wheel odometry with Kalman filtering to reduce localization error to 0.5%.
- Presented the work as an ICROS 2023 workshop paper and received the ICROS 2023 Undergraduate Best Paper Award.

ERIC_A: Indoor Autonomous Delivery Robot *Hanyang University*

Project Lead *Sep. 2022 - Dec. 2022*

- Led an 8-member indoor delivery robot project integrating web order dispatch, ROS navigation, and YOLO-based human detection.
- Improved odometry error from 5–10% to 1% through IMU/encoder Kalman-filter fusion.
- Achieved 30 mm $\pm$ 1 destination accuracy at 0.5 m/s average robot speed.

Honors & Awards

Domestic Awards

Nov. 2023	**IITP President Award**, SW Talent Festival (National Capstone Competition)	*South Korea*
Jun. 2023	**Top Award**, Intelligent Robot AI Idea Competition	*South Korea*
Aug. 2022	**2nd Place**, Ansan Young Innovators Design Thinking Workshop	*South Korea*
Jul. 2022	**1st Prize**, Cancer Patient Care and Health Product Idea Competition	*South Korea*
Aug. 2021	**1st Prize**, ODA Idea Competition	*South Korea*

Teaching & Service

IEEP Tanzania International Science Volunteer *Arusha, Tanzania*

Volunteer *Jan. 2024*

- Participated in an international science volunteer program.
- Contributed to science education.

HYMEC - Robot Making Club *Ansan, South Korea*

Management Lead / Club President *Jan. 2021 - Dec. 2022*

- Served as management lead and later club president for the robotics club.
- Organized mobile-robot mission hackathons and peer robotics activities.

Tutoring Mentor *Ansan, South Korea*

- Mentored students in Git, ROS/ROS2, and Docker.

Technical Skills

Programming	C/C++, Python
Robotics	ROS2, ROS1, MoveIt2, motion planning, collision detection
Perception & Geometry	PointCloud2 processing, Open3D
Tools	Git, Docker, Ray, PyTorch, pytest, mypy, ruff, uv
Robot Platforms	UR e-Series, RB20, TM20, HDR20-17, mobile robots